# On the Boundary Between Theory and Practice: Reflections on the SVP-200 Challenge

Jintai Ding

Xi'an Jiaotong-Liverpool University
Basque Center for Applied Mathematics

## Background – Public Key Cryptography

- Motivation of PKC – security and authentication needs of large compouter networks
- PKC - a revolutionary idea: **One to Many**
- Public key and private key pair: Public key **(ONE)** can be used by **Many and anyone** !!!
- The theoretical foundation: one way functions:

$$N = P \times Q$$

- Our current digital economy depends on two hard mathematical problems: Integer Factorization and Discrte Logrithm.

# Background – Post-Quantum Cryptography

- PQC – PKC that could resist quantum computer attacks
  Shor's algorithm in 1994, Chuang's implementation in 2000 to fcator 15.
- NIST started the PQC standarization in 2016
- New mathematical problems.
  Lattice, Multivariate Polynomials, Hash functions, Error Correction Codes

## Background- Post-Quantum Cryptography

Selected NIST Post-quantum Algorithms 2022:

- **CRYSTALS-KYBER** (module LWE)
- **CRYSTALS-DILITHIUM** (module LWE)
- **FALCON** (NTRU lattice)
- **SPHINCS+** (non-lattice)

The view: lattice-based schemes are the most attractive candidates for their compactness, efficiency, and provable security.

## Lattice

**Definition**

A **lattice** (denoted by $\mathcal{L}$) is a discrete subgroup in $\mathbb{R}^m$.

**Proposition**

*$\mathcal{L}$ always admits an* **integral basis** $\mathbf{B} = (\mathbf{b}_0, \mathbf{b}_1, \cdots, \mathbf{b}_{n-1})$ *such that*

$$\mathcal{L} = \{\lambda_0\mathbf{b}_0 + \lambda_1\mathbf{b}_1 + \cdots + \lambda_{n-1}\mathbf{b}_{n-1} : \lambda_i \in \mathbb{Z},\ i = 0, \cdots, n-1\}.$$

A lattice has infinitely many bases, each related to another by a unimodular transformation.

## Post-Quantum Cryptography – Lattice

- The first promising lattice cryptosystem: NTRU – From Ring to Lattice (Brown!)
- Then LWE: the learning wirth error problem
- The basic idea:
  1. a good basis with shortest vector as the secret key
  2. a "bad" or random basis as the public key

  The shortest vector problem – SVP
  The approximate shortest vector problem – ASVP
- Ajtai: Average to Worst case reduction
  Provable security

# Post-Quantum Cryptography – Lattice

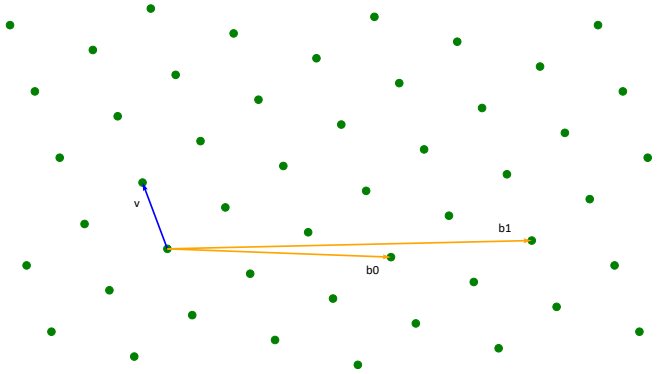
1. In two dimensional space:
   $V_1 = (9, 16)$,
   $V_2 = (5, 9)$
   What are the shortest vectors?

# Post-Quantum Cryptography – Lattice

2. $(0,1) = (-5) \times (9,16) + 9 \times (5,9)$
   $(1,0) = 9 \times (9,16) + (-16) \times (5,9)$

# Lattice

**Lattice reduction** refers to finding a "good" basis from an arbitrary one.



## SVP and CVP

We denote the length of the shortest non-zero vector in $\mathcal{L}$ by $\lambda_1(\mathcal{L})$.

**Definition (Shortest Vector Problem $\mathrm{SVP}_\gamma$)**

$\gamma = \gamma(n)$ is a function of $n$. Given a basis $\mathbf{B} \in \mathbb{Q}^{m \times n}$ of $\mathcal{L}$, output a non-zero lattice vector $\mathbf{v}$ such that $\|\mathbf{v}\| \leq \gamma(n)\lambda_1(\mathcal{L})$.

**Definition (Closest Vector Problem $\mathrm{CVP}_\gamma$)**

$\gamma = \gamma(n)$ is a function of $n$. Given a basis $\mathbf{B} \in \mathbb{Q}^{m \times n}$ of $\mathcal{L}$ and a target vector $\mathbf{t} \in \mathbb{Q}^m$, output a lattice vector $\mathbf{v}$ such that $\|\mathbf{v} - \mathbf{t}\| \leq \gamma(n)\,\mathrm{dist}(\mathbf{t}, \mathcal{L})$.

## Gaussian Heuristic

The **determinant** of $\mathcal{L}$ is $\det(\mathcal{L}) = \sqrt{\det(\mathbf{B}^T\mathbf{B})}$. While finding a vector of length $\lambda_1(\mathcal{L})$ is generally hard, for "random" lattices, the **Gaussian Heuristic** provides a good estimate for $\lambda_1(\mathcal{L})$:

$$\mathrm{gh}(\mathcal{L}) = \frac{\Gamma(\frac{n}{2}+1)^{1/n}}{\sqrt{\pi}} \cdot (\det(\mathcal{L}))^{1/n} \approx \sqrt{\frac{n}{2\pi e}} \cdot (\det(\mathcal{L}))^{1/n}.$$

**Remark**

*The notion of a "random" lattice here is subtle. A necessary condition is that the given basis and its dual should not, due to its construction, contain unusually short vectors.*

## Background – Provable Security

Provablely secure?

- **Provable security** of the lattice schemes has very strict requirement on the parameters!
- None of the lattice schemes' paramters satisfy the provable secure requirement!
- Then we need to play the games as usual: the pratical security. (But still claim provable secure?)
- So how hard are the lattice attacks? Do we have both theoretical and pratical support?

# Background – How hard is the lattice reduction problem?

- In general, SVP is NP-hard.
- Approximate SVP (LWE) with polynomial factors?
- Then again, let us look at practical security in terms of concrete attacks?

# Background – How hard is the lattice reduction problem?

- The first break through – LLL
- Approximation Factor: $2^n$, running time polynomial!
- the length of the output of the shortest vector $= (RHF)^n \lambda_1$
  $\lambda_1 =$ the length of the SV.
- Root Hermite Factor: Theory: **1.075**; Practice **1.02** ! Why?
  Sand pile model but .....
- BKZ: call SVP subroutine; Theory? ....
  Practical SVP?

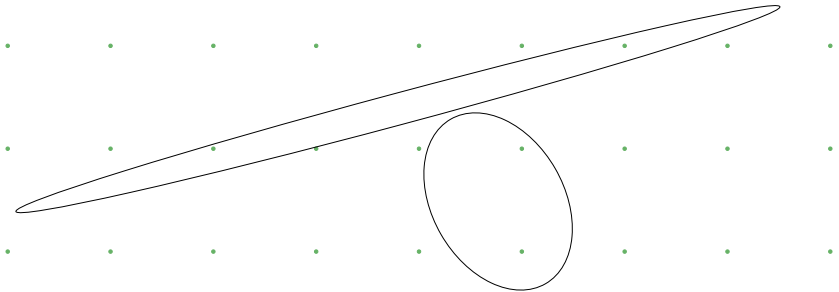
## Lattice Reduction

**Goal**

To find a "good" basis for the lattice, given an arbitrary one.

**Question**

What is a "good" basis?



## Size Reduction

Intuitively, we want the basis vectors to be short and nearly orthogonal.

$$
\begin{aligned}
\mathbf{b}_0 &= \mathbf{b}_0^* \\
\mathbf{b}_1 &= \mu_{1,0}\mathbf{b}_0^* + \mathbf{b}_1^* \\
\mathbf{b}_2 &= \mu_{2,0}\mathbf{b}_0^* + \mu_{2,1}\mathbf{b}_1^* + \mathbf{b}_2^* \\
&\cdots \\
\mathbf{b}_{n-1} &= \mu_{n-1,0}\mathbf{b}_0^* + \cdots + \mu_{n-1,n-2}\mathbf{b}_{n-2}^* + \mathbf{b}_{n-1}^*
\end{aligned}
$$

**Size Reduction**

It's straightforward to make all $\mu_{i,j} \in [-0.5, 0.5)$.

## Local Projected Lattice

Besides, we may quantify the quality of a basis using the **Potential**:

$$\mathrm{Pot}(\mathbf{B}) = \prod_{k=0}^{n-1} \|\mathbf{b}_k^*\|^{2(n-1-k)}$$

### Definition (Local Projected Lattice)

For $0 \leqslant i < j \leqslant n$, the local projected basis $\mathbf{B}_{[i,j]}$ of $\mathbf{B}$ is the projection of $(\mathbf{b}_i, \ldots, \mathbf{b}_{j-1})$ onto $\mathrm{span}(\mathbf{b}_0, \ldots, \mathbf{b}_{i-1})^{\perp}$:

$$\mathbf{B}_{[i,j]} = \left( \mathbf{b}_i^*, \mu_{i+1,i}\mathbf{b}_i^* + \mathbf{b}_{i+1}^*, \ldots, \sum_{k=i}^{j-2} \mu_{j-1,k}\mathbf{b}_k^* + \mathbf{b}_{j-1}^* \right)$$

We denote by $\mathcal{L}_{[i,j]}$ the **local projected lattice** generated by $\mathbf{B}_{[i,j]}$.

## LLL Algorithm

The celebrated Lenstra-Lenstra-Lovász algorithm from a high-level view:

```
while (swapping some b_i and b_{i-1} reduces Pot by 1%) {
    swap b_i and b_{i-1} and size reduce;
}
```

- Provable to stop in polynomial time, ensuring $\|\mathbf{b}_0\| \leq 1.075^n \cdot \det(\mathcal{L})^{1/n}$.
- In practice: Not fast, but usually achieves $\|\mathbf{b}_0\| \approx 1.02^n \cdot \det(\mathcal{L})^{1/n}$.
- It cannot make the basis worse; the running time depends on how much the quality is improved. Always worth performing as a preprocessing step.

## BKZ Algorithm

The Block Korkine-Zolotarev algorithm tries to ensure

$$\lambda_1(\mathcal{L}_{[i,\min\{i+\beta,n\}]}) = \|\mathbf{b}_i^*\|, \quad i = 0, \ldots, n-1$$

by calling SVP oracles on local projected lattices of blocksize $\beta$ in some order.

- Behavior not really provable but practically predictable.
- Provides trade-off: $2^{\tilde{O}(n^c)}$ time for a $2^{\tilde{O}(n^{1-c})}$ approximation factor.

The best known method for approx-SVP is BKZ with a sieving SVP oracle.

## Background – SVP Enumeration

- Given a basis $B$ of a lattice, determine a region $R$, such that the SV is in $R$. Enumerate all the points in $R$
- How many lattice vectors needed in $R$: $2^{O(n \log n)}$, space small
  Pruning ....
  The best player for a while! ( Till 2018? )

## Background – Sieving – the Usurper

- Sample $2^{\alpha n}$ points, whose length is less or equal to $r_0$ therefor inside a sphere of radius $r_0$.
- We repeatedly use a "Sieving" procedure to shrink the size of the spheres to find the SV.

## Background

"our best estimate for the gate cost of attacking Kyber512 using known techniques is about $2^{147}$ (or $2^{145}$..."

"Combining the above estimates of **the cost of memory access**...the realistic cost of attacking Kyber512 is the equivalent of about $2^{160}$ bit operations/ gates" [1]

**Is it really true?**

[1] https://csrc.nist.gov/csrc/media/Projects/post-quantum-cryptography/documents/faq/Kyber-512-FAQ.pdf

## Background: our work

Provable security means breaking these lattice-based schemes is not easier than solving some module/ring-LWE problems.

Hardness estimation of structured lattice problems:

- Expoliting the algebraic structure, e.g. there exists[2] quantum polynomial time algorithm for $2^{\tilde{\mathcal{O}}(\sqrt{n})}$-ideal SVP.
- General Lattice reduction algorithms, BKZ with sieving-based SVP oracle.

[2] [CDW21] Ronald Cramer, Léo Ducas, Benjamin Wesolowski. Mildly Short Vectors in Cyclotomic Ideal Lattices in Quantum Polynomial Time.

## Background

We focus on the complexity of SVP oracles in BKZ. Progress on SVP in last decade[3]:

- 2013, SVP130 by Kenji Kashiwabara and Masaharu Fukase (RSR)
- 2015, SVP140 by Kenji Kashiwabara and Tadanori Teruya (RSR with small cluster)
- 2017, SVP150 by Kenji Kashiwabara and Tadanori Teruya (RSR with small clusters)
- 2018, SVP155 by M. Albrecht, L. Ducas, G. Herold, E. Kirshanova, E. Postlethwaite, M. Stevens, P. Karpman (bgj1 sieve)
- 2020, SVP170 by L. Ducas, M. Stevens, W. van Woerden (HK17[4]-sieve with GPU)
- 2021, SVP180 by L. Ducas, M. Stevens, W. van Woerden (HK17-sieve with GPU)

[3] see https://www.latticechallenge.org/svp-challenge/halloffame.php

[4] Herold, G., Kirshanova, E.: Improved Algorithms for the Approximate k-list Problem in Euclidean Norm.

## Background

Sieving has finally won over enumeration, becoming the benchmark for assessing SVP hardness.

| | |
|---|---|
| Gauss sieve / NV sieve | $2^{0.415n+o(n)}$ |
| BGJ14 | $2^{0.377n+o(n)}$ |
| HK17 | $2^{0.349n+o(n)}$ |
| Laa15a | $2^{0.337n+o(n)}$ |
| BGJ15 | $2^{0.311n+o(n)}$ |
| LdW15 | $2^{0.297n+o(n)}$ |
| BDGL16 | $2^{0.292n+o(n)}$ |

Impossible to attain an exponent less than $\frac{1}{2}\log_2\left(\frac{3}{2}\right) \approx 0.292$ by better NNS strategy.[5]

[5] Kirshanova, E., Laarhoven, T.: Lower bounds on lattice sieving and information set decoding.

## Background

Improvements in the $o(n)$ term contribute to the main part of last decade's (practical) progress in SVP:

- XOR-POPCNT trick (simhash), Duc18[6], FBB$^+$15[7], Cha02[8]
- progressive sieving, Duc18, LM18[9]
- preprocessing (SubSieve, G6K) Duc18, ADH$^+$19[10]
- Dimensions for Free, Duc18, ADH$^+$19

[6] [Duc18] Léo Ducas, Shortest vector from lattice sieving: A few dimensions for free.

[7] [FBB$^+$15] Robert Fitzpatrick, Christian H. Bischof, Johannes Buchmann, Özgür Dagdelen, Florian Göpfert, Artur Mariano, and Bo-Yin Yang, Tuning GaussSieve for speed.

[8] [Cha02] Moses Charikar, Similarity estimation techniques from rounding algorithms.

[9] [LM18] Thijs Laarhoven and Artur Mariano, Progressive lattice sieving

[10] [ADH$^+$19] Albrecht, M.R., Ducas, L., Herold, G., Kirshanova, E., Postlethwaite, E.W., Stevens, M.: The general sieve kernel and new records in lattice reduction.

## Background

"our best estimate for the gate cost of attacking Kyber512 using known techniques is about $2^{147}$ (or $2^{145}$..."

"Combining the above estimates of **the cost of memory access**...the realistic cost of attacking Kyber512 is the equivalent of about $2^{160}$ bit operations/ gates" [11]

[11] https://csrc.nist.gov/csrc/media/Projects/post-quantum-cryptography/documents/faq/Kyber-512-FAQ.pdf

## Background

The main obstacle for solving larger SVP instances is the random memory access cost:

- SVP200 will require about 10 terabytes of memory.
- The limited host-device bandwidth significantly hampers the BDGL sieve's practical performance.

The issue is becoming more severe as the sieving dimension increases.

## Background

Randomly access to a ($N$ bits) 3D massive storage array costs $\mathcal{O}(N^{1/3})$.
A list of energy consumption for different operations:[12]

| Operation | Energy (pJ) |
|---|---|
| 8b Add | 0.03 |
| 16b FB Add | 0.4 |
| 32b FB Add | 0.9 |
| 8b Mul | 0.2 |
| 16b FB Mult | 1.1 |
| 32b FB Mult | 3.7 |
| 32b SRAM Read (8KB) | 5 |
| 32b DRAM Read | 640 |

[12] Computer Architecture: A Quantitative Approach, 6th Edition, P29, Energy numbers are from Mark Horowitz *Computing's Energy problem (and what we can do about it)*. ISSCC 2014

## Background

This work suggests the inherent structure of BGJ sieve is significantly more memory-efficient than BDGL sieve.

- $2^{0.2075n+o(n)}$ streamed main memory accesses is enough
- time complexity of a modifed BGJ sieve is very close to BDGL sieve, faster for all practical dimensions.
- supported by implementation.

# Background

Table: Comparison with Previous GPU Records

| Dim | Walltime | Platform | FLOP |
|---|---|---|---|
| 179 | $11.2d$ | 112 cores, no GPU* | $2^{66.0} \approx 2^{13.6} \cdot (3/2)^{179/2}$ int8 op |
| 183 | $30d$ | 112 cores, no GPU* | $2^{67.4} \approx 2^{13.9} \cdot (3/2)^{183/2}$ int8 op |
| 180 | $51.6d$ | 4 $\times$ Nvidia RTX 2080ti | $2^{69.9} \approx 2^{17.3} \cdot (3/2)^{180/2}$ fp16 op$^{\dagger}$ |
| 186 | $50.3d$ | 4 $\times$ Nvidia A100 | $2^{71.4} \approx 2^{17.0} \cdot (3/2)^{186/2}$ fp16 op$^{\ddagger}$ |

[12] † See Table 1 in [DSvW21] for more details.

## Recall

$\mathcal{L}$ is an $n$-dimensional lattice with basis $\mathbf{b}_1, \cdots, \mathbf{b}_n$.
The Gram-Schmidt orthogonalization: $\mathbf{b}_1^*, \cdots, \mathbf{b}_n^*$.
The dual basis: $\mathbf{b}_1^\vee, \cdots, \mathbf{b}_n^\vee$.
Gaussian heuristic: $\mathrm{gh}(\mathcal{L})$

## Recall

Sieving, a natural generalization of Gauss's algorithm

- Maintain a list of lattice vectors
- Find "Reducing-pairs" which generate new short vectors
- Replace the longest vectors with the short ones

| | |
|---|---|
| Initialize? | Arbitrarily sample $N = 3.0 \cdot (4/3)^{n/2}$ vectors |
| Stop? | When the list saturates the ball of radius $\sqrt{4/3} \cdot \mathrm{gh}(\mathcal{L})$ |
| Reducing-triples? | |

## Recall

The cost of sieving is dominated by the search for reducing-pairs.

Accelerating the Nearest Neighbor Search (NNS) on high-dimensional sphere by locality-sensitive filter:

- Put the list vectors into buckets
- Search for reducing pairs in each buckets (naively)
- Reducing-pairs are more likely to be in the same bucket

## Recall

**Definition (Spherical cap shaped filters)**

A vector $\mathbf{v}$ can pass the filter $\mathcal{F}_{\mathbf{c},\alpha}$ with center $\mathbf{c}$ and radius $\alpha$ if and only if $|\langle \mathbf{v}, \mathbf{c} \rangle| \geqslant \alpha \|\mathbf{v}\| \|\mathbf{c}\|$.

- BDGL sieve uses single layer of spherical cap shaped filters
- Asymptotically $\alpha \to 0.5$, $2^{0.292n+o(n)}$ buckets, each of size $2^{o(n)}$
- Bucket center $\mathbf{c}$ comes from random product code for quick bucketing[13]
- The slow down caused by non-uniformness of $\mathbf{c}$ is subexponential[14]

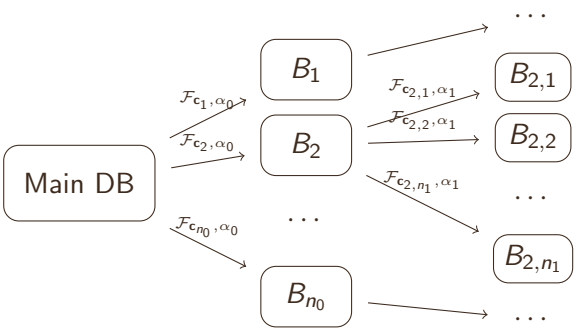
[13] [BDGL16] Anja Becker, Léo Ducas, Nicolas Gama, and Thijs Laarhoven, New directions in nearest neighbor searching with applications to lattice sieving.

[14] [Duc22] Ducas, L.: Estimating the hidden overheads in the bdgl lattice sieving algorithm.

## Recall

### BGJ15

Generates progressively smaller buckets by applying a series of random filters to the main database.




pairwise search here

...

## Time Complexity

Here, we replace the original filters with spherical cap-shaped filters.

- These filters have been shown to be optimal in terms of time complexity
- bgj1 sieve in [ADH+19] has proven efficient in practice.

## Time Complexity

Let $P_f$ be the probability that a vector will pass a random filter from $\mathcal{F}$, and $P_p$ is the probability that a pair of vectors, which form an angle of $\pi/3$, are both accepted by the same random filter.

**Theorem (Complexity of AllPairSearch-BGJ15[15])**

*Suppose $L$ is a list of $N$ uniformly random vectors in the sphere of dimension $n$, $\rho$ is the exponent such that $P_f^\rho = P_p$, then the time complexity is $\tilde{\mathcal{O}}(N^\rho)$.*

[15] [BGJ15] Becker, A., Gama, N., Joux, A.: Speeding-up lattice sieving without increasing the memory, using sub-quadratic nearest neighbor search.

## Time Complexity

spherical caps $\mathcal{C}_{\mathbf{c},\alpha} = \{\mathbf{x} \in \mathbb{R}^n | \|\mathbf{x}\|^2 = 1, \langle \mathbf{x}, \mathbf{c} \rangle \geqslant \alpha \|\mathbf{c}\|\}$.
wedges (i.e. intersections of spherical caps) $\mathcal{W}_{\mathbf{c}_1,\alpha_1,\mathbf{c}_2,\alpha_2} = \mathcal{C}_{\mathbf{c}_1,\alpha_1} \cap \mathcal{C}_{\mathbf{c}_2,\alpha_2}$.

### Lemma (Volume of spherical caps and wedges[16])

*Let $\mu$ be the canonical Lebesgue measure, $\mathcal{S}^{n-1}$ be the unit sphere in $\mathbb{R}^n$, then for any $\alpha \in (0,1)$ we have*

$$\frac{\mu(\mathcal{C}_{\mathbf{c},\alpha})}{\mu(\mathcal{S}^{n-1})} = \mathrm{poly}(n) \cdot \left(\sqrt{1-\alpha^2}\right)^n.$$

*Furthermore, if the angle between $\mathbf{c}_1$ and $\mathbf{c}_2$ is $\theta$, then*

$$\frac{\mu(\mathcal{W}_{\mathbf{c}_1,\alpha,\mathbf{c}_2,\alpha})}{\mu(\mathcal{S}^{n-1})} = \mathrm{poly}(n) \cdot \left(\sqrt{1-\frac{2\alpha^2}{1+\cos\theta}}\right)^n.$$

[16] [BDGL16] Anja Becker, Léo Ducas, Nicolas Gama, and Thijs Laarhoven, New directions in nearest neighbor searching with applications to lattice sieving.

## Time Complexity

$P_f = \text{poly}(n) \cdot (1-\alpha^2)^{n/2}$ and $P_p = \text{poly}(n) \cdot (1-\frac{4}{3}\alpha^2)^{n/2}$. This implies that asymptotically

$$\rho \approx \ln(1-\frac{4}{3}\alpha^2)/\ln(1-\alpha^2)$$

- For original BGJ sieve, $\rho = 1.5$
- Achieves the asymptotically optimal time complexity of $\tilde{\mathcal{O}}(N^{4/3})$, when the dataset is sparse ($N = 2^{o(n)}$)

## Time Complexity

- It's not guaranteed that filters optimal in the sparse regime will remain optimal in the dense regime ($N = 2^{\mathcal{O}(n)}$).
- Cross-polytope hashing is known to be optimal in the sparse case[17], but it leads to a suboptimal time complexity of $2^{0.297n+o(n)}$ when applied to lattice sieving[18].
- $2^{(0.297-0.292)\cdot(380-140)} = 2^{1.2}$, does it really matters?

[17] [AIL+15] Andoni, A., Indyk, P., Laarhoven, T., Razenshteyn, I., Schmidt, L.: Practical and optimal lsh for angular distance.

[18] [LdW15] Laarhoven, T., Weger, B.: Faster sieving for shortest lattice vectors using spherical locality-sensitive hashing

## Memory Access Cost

**Observation**

The bucket size decreases by several orders of magnitude after each filter, allowing the sub-buckets to be stored in a much smaller, and therefore faster, storage device.

- No communication between these sub-buckets is necessary.
- Data movement is streamed, except for the filtering stage.

# Memory Access Cost

Supported by implementation, Sieving dimension $= 140$

Table: Profiling Data of bgj3-amx

| Step | Filter-0 | Filter-1 | Filter-2 | Reducing |
|---|---|---|---|---|
| Speed (TOPS) | 11.81 | 11.10 | 39.19 | 116.4 |
| Bucket size | 278.8GB | 3.386GB | 80.75MB | 556.7KB |
| Data in | RAM | RAM | L3-Cache | L2-Cache |
| Total Time | 544.7s | 451.4s | 762.4s | 3397s |

## Memory Access Cost

- BDGL uses a single filter layer to generate $2^{\mathcal{O}(n)}$ buckets: randomly write to exponentially large space.
- If BGJ uses $\mathcal{O}(\log(n))$ successive filter layers, the number of subbuckets for each bucket can be $2^{\mathcal{O}(n/\log(n))}$.
- Practical value of subbuckets: $\sim 2^7$ for sieving 140.

## Memory Access Cost

- The computations required are superlinear (with an exponent range from $0.292/0.2075 \approx 1.41$ to 2) in the size of the subbucket.
- Streamed memory movement of $N$ bits costs $\mathcal{O}(N^{4/3})$.
- As long as the subbucket size is larger than some constant, the memory access overhead, caused by the streamed memory access is negligible.

## Memory Access Cost

**For last several layer of buckets?**

Not observed in today's computational architectures, but it may matter for serious attacks using ASICs.

How to check for duplicates before a vector is inserted into the database without a UidHashTable?
Sort the list vectors and the newly found short vectors together, and then remove the duplicates and the longest ones.

$$\mathcal{O}(2^{0.2075n+o(n)} \log(2^{0.2075n+o(n)})) = 2^{0.2075n+o(n)}$$

## Implementation Details

**Fact**

Lattice reduction and sieving are both numerically unstable.

The lattice basis $\mathbf{b}_1, \cdots, \mathbf{b}_n$ and the Gram-Schmidt orthogonalization $\mathbf{b}_1^*, \cdots, \mathbf{b}_n^*$ are represented by quad_float, with precision of 106 bits.

- 53 bits "double" may fail.
- We have vectorized the quad_float operations.

## Implementation Details

The sieving vectors are rounded to 8 bit integers.

- save half of the bandwidth and computation resources.
- reduce the main database size by 40%.
- 4 bit is not enough.

Scale the entries by $254.0 \cdot (\sup_{1 \leqslant i \leqslant n} \|\mathbf{b}_i^*\|)^{-1}$ then round: After size reduction, overflow may not happen.

## Implementation Details

Each vector, before inserted into the database, should carefully checked and normalized.
Normalization means: recover the integer coefficients wrt the local projected basis, and then compute fp32 vectors and finally round.
Dual basis of a well-reduced basis can be quite long.

**Fact**

Even if the sieving vectors are represented by "double", the normalization is still necessary.

## Implementation Details

Choice of filtering batchsize and filter radius?

**Example**

For bgj1 the asymptotically optimal choice ($\alpha_0 = 0.366$) can be far from the practical optimum ($\alpha_0 = 0.315 \sim 0.325$)

Table: Chosen Filter Radius in bgj1, bgj3, bgj3, and bgj3-amx

| Algorithm | $\alpha_0$ | $\alpha_1$ | $\alpha_2$ |
|---|---|---|---|
| bgj1 | 0.325 | - | - |
| bgj2 | 0.257 | 0.280 | - |
| bgj3 | 0.200 | 0.210 | 0.280 |
| bgj3-amx | 0.210 | 0.215 | 0.285 |

## Implementation Details

XOR-POPCNT trick (simhash) is disabled in our implementation.

- the theoretical throughput for a dot product is less than 4 clock cycles
- simhash check makes the memory access pattern unpredictable
- additional 32 bytes memory per vector

## Implementation Details

How do we compute dot products?

- "avx2" implementation: 8 dot products in parallel, first computed by vpdpbusd on ymm registers, then horizontally add the 32-bit results by vphaddd instruction.
- "amx" implementation: compute 256 dot products by only 3 tdpbssd instructions, transpose one of the 16 by 64 int8_t matrices before loaded into tmm registers by vpunpckldq, vpunpckhdq, and vshufi64x2.
- do not collect reducing-triples to avoid additional comparisons.

## More

An implementation of BGJ sieve, with the sieving database on SSD and computation on multi-GPU, was done several months ago.

- Progressive sieve spends about 6 minutes and 8 hours on dimensions 140 and 160
- Set new lattice record — 200 dimensional Darmstadt SVP challenge

**Fact**

The community can now solve about 1,000,000 × harder SVP instances than ten years ago!

# Our record: Joint work with Ziyu Zhao

2025.03

version of the generator.

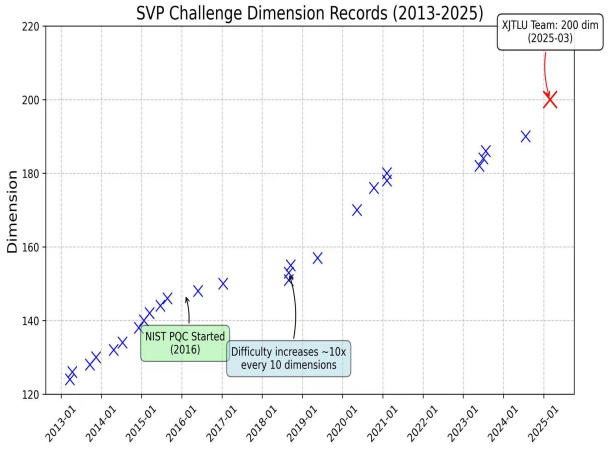
HALL OF FAME

| Position | Dimension | Euclidean norm | Seed | Contestant | Solution |
|---|---|---|---|---|---|
| 1 | 200 | 3723 | 0 | Ziyu Zhao, Jintai Ding | vec |

10 Terra byte of memeory in harddisk

# Our record: Joint work with Ziyu Zhao

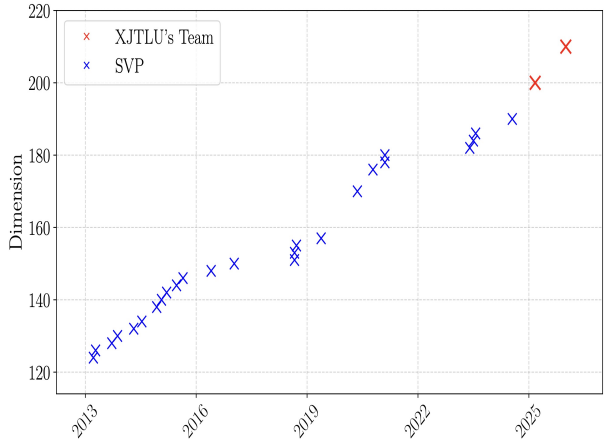
2025.03




Breaking Kyber 512 – SVP 400!

# Our new record: Joint work with Ziyu Zhao

2026.02




Breaking Kyber 512 – SVP 400!

## Refined Security Analysis

Does ... takes less than $2^{143}$ gates?
Does ... easier than breaking AES128?
Uncertainty exists regarding:

- the exact time complexity
- the memory access cost for the final bucket layers
- theory?

# Thank you

Questions?